

Hands on Homework #7 (Center of Gravity)

EGN 3310 - 0M01 Engineering Analysis: STATICS



MWF, 11:30AM-12:20PM

Submitted by: Group 38

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Hands-On Method (Physical Model)

Photos of our cut out models

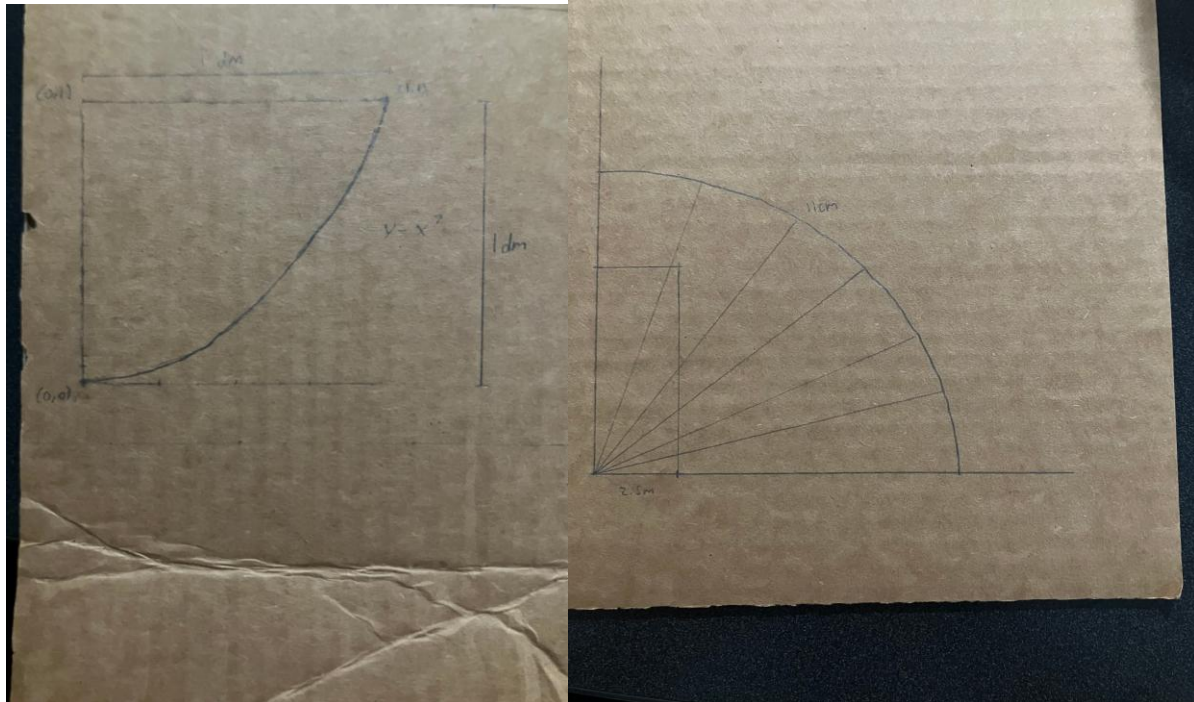


Fig. 1.01:

Drawing of shape with Integrals Method (before cut)

Fig. 1.02:

Drawing of shape with Composite Areas Method (before cut)

Graphical Method:

To find the centroid of an object without calculations, you have to hang it from at least 2 different axes, then mark where the 2 lines of intersection intersect at.

Below is the proof as well as the results:



Fig. 1.03:

Shape defined by a quarter of a circle and small rectangle's area subtracted; finding centroids by hanging from different axes

Fig. 1.04:

Shape defined by $y = x^2$; finding centroids by hanging from different axes

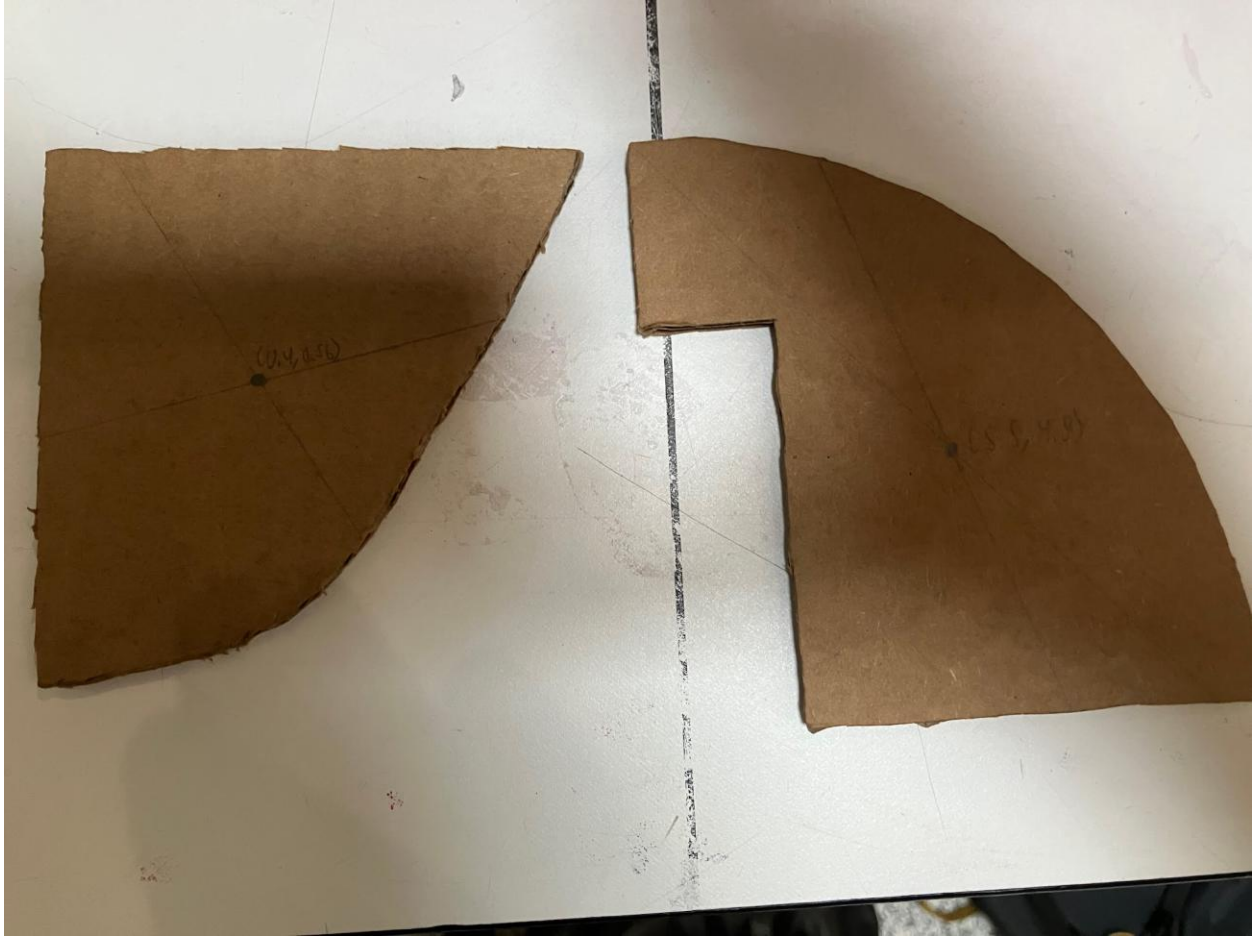
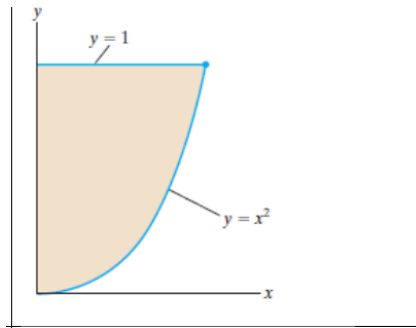


Fig: 1.05: graphical result of centroids of both shapes. The figure on the left had a centroid at $(0.4, 0.56)$ while the figure on the right had a centroid at $(5.5, 4.9)$

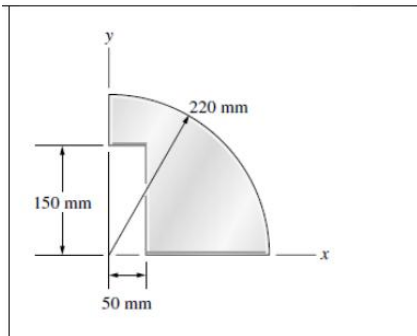
Analytical

The hands-on portion for this was very easy this week. We simply had to find the centroid of the cut out models, which we accurately cut out.

To find the centroid, we simply had to hang each shape from two opposite ends and draw lines through the whole shape. The point at which they intersect is the centroid.



For this shape, we found that the centroid was at $x=0.4\text{dm}$ and $y=0.56\text{dm}$

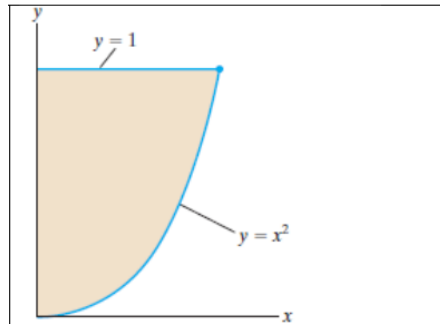


For this shape we found that the centroid was at $x=5.5\text{cm}$ and $y=4.9\text{cm}$

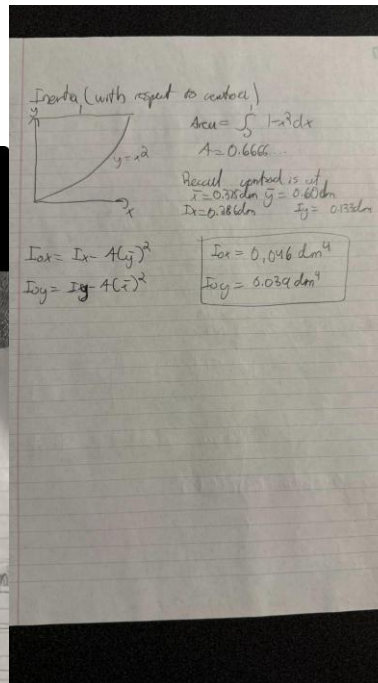
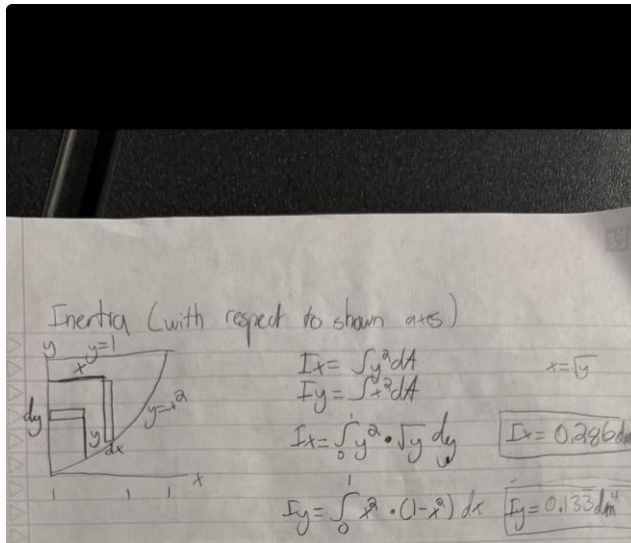
That concludes this section. It was quite short since all we had to find was the centroid, which is quite simple.

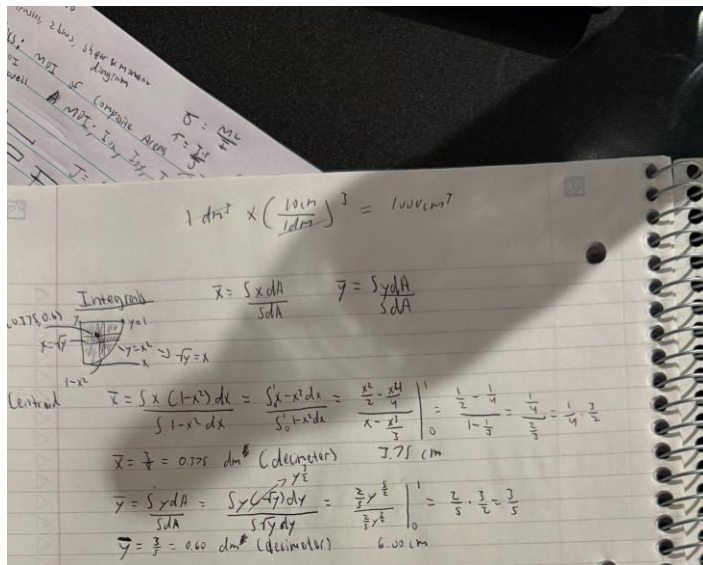
Hand Calculation (Theoretical Method)

Integral version



Paper with all our calculations:





For the integral part of our question, we had to solve for centroid, inertia with respect to axes and centroid.

For this one, we set the units as decimeters for our convenience.

Centroid

To find the centroid, we used the formulas below to find the centroids in the x and y directions respectively:

$$x = \int x^2 dA / \int dA \quad y = \int y dA / \int dA$$

Centroid of x:

$$x = \int x dA / \int dA = \int x(1 - x^2) dx / \int (1 - x^2) dx = 0.375 \text{ dm}$$

Centroid of y:

$$y = \int y dA / \int dA = \int y(y^{1/2}) dy / \int y^{1/2} dy = 0.60 \text{ dm}$$

Inertia (with respect to axes)

So to find the inertia with respect to axes, we had to use two simple formulas.

$$I_y = \int x^2 dA \text{ and } I_x = \int y^2 dA$$

Per usual, the hard part about this is finding dA .

For the dA for the y -equation, we found it the following way.

We drew the rectangle dy perpendicular to the y line.

We then found dy by finding the area in terms of x , we we found was $1 - x^2$

This gave us the final equation $I_y = \int_0^1 x^2(1 - x^2)dx$, and by inputting it into the calculator we got got that $I_y = 0.133 \text{ dm}^4$

For the dA for the x -equation, we found it the following way.

We drew the rectangle dy perpendicular to the x line.

We then found dy by finding the area in terms of y , we we found was $\sqrt{y}$

This gave us the final equation $I_x = \int_0^1 y^2(\sqrt{y})dy$, and by inputting it into the calculator we got got that $I_x = 0.286 \text{ dm}^4$

Inertia (with respect to centroid)

Now that we have the inertia with respect to axes, as well as the coordinates of the centroid, finding the inertia with respect to the centroid is much easier.

First we calculated the area of the shape using a basic integral:

$$\text{Area} = \int_0^1 1 - x^2 dx = 0.666\dots$$

Now recall that the centroid is at $x=0.375\text{dm}$ and $y= 0.60\text{dm}$.

Also recall that the inertial with respect to the axes are $I_x = 0.286\text{dm}$ and $I_y = 0.133\text{dm}$

Using these values, we can easily find the inertia from the centroid using the following formulas:

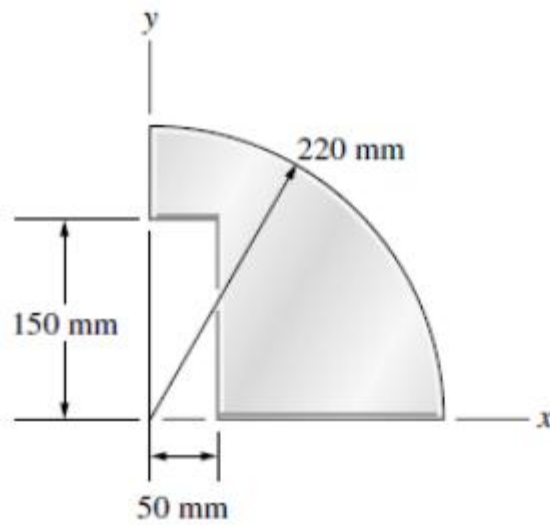
$$I_{ox} = I_x - A(\underline{y})^2$$

$$I_{oy} = I_y - A(\underline{x})^2$$

Then solving for those we get that:

$$I_{ox} = 0.046\text{dm}^4$$

$$I_{oy} = 0.286\text{dm}^4$$



Shape version

Paper with all our calculations:

Figure A

	A	x	y	Ax	Ay
Δ	$\frac{\pi(0)^2}{4} = 95.03$	4.67	4.67	443.79	443.79
$\square$	18.75	1.25	3.75	23.44	70.31
Σ	76.28			420.35	373.48
$\bar{x}$	$\frac{420.35}{76.28} = 5.51$				
$\bar{y}$			$\frac{373.48}{76.28} = 4.90$		

$\bar{x} = 5.51 \text{ cm}$ $\bar{y} = 4.90 \text{ cm}$

Inertia (with respect to centroid)

Recall that $\bar{x} = 5.51 \text{ cm}$ $\bar{y} = 4.90 \text{ cm}$

$I_x = I_{x0} + A(\bar{y})^2$ $I_y = I_{y0} + A(\bar{x})^2$

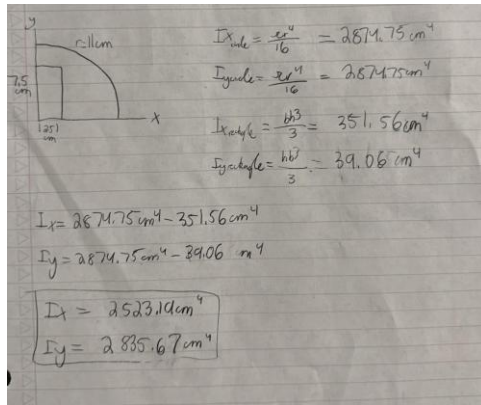
$I_{x0} = 2523.14 \text{ cm}^4$ $I_{y0} = 2135.67 \text{ cm}^4$

Figure	Area	D_x	D_y	$A D_x^2$	$A D_y^2$
Δ	95.03	0.84	-0.23	67.05	5.03
$\square$	18.75	4.26	1.15	340.27	24.80
Σ	76.28			407.32	29.83

$I_x = 2523.14 + 407.32 = 2930.46 \text{ cm}^4$

$I_y = 2135.67 + 29.83 = 2165.50 \text{ cm}^4$

We do not have to use these



For the shape part of our question, we had to solve for the centroid, inertia with respect to axes and centroid.

For our convenience, we converted all the measurements to centimeters and then divided them by two.

Centroid

When it comes to calculating the centroid, we first started off by forming a graph based on the figure shape, area, x and y length, as well as its area time by the x and y.

Using these six we first started off by defining all shapes and their values provided:

Figure	Area	x	y	A * x	A*y
Quarter Circle	95.03	4.67	4.67	443.79	443.79
Rectangle	18.75	1.25	3.75	23.44	70.31

After we determine all the values, we take the sum of the area and the sum of both values of Ax and Ay. Keep in mind for our rectangle, since it was a shape that cut into the shape of the quarter circle we must subtract its value from the

quarter circle. Thus, our value would be as so:

Sum of Area =	Sum of Ax =	Sum of Ay =
76.28	420.35	373.48

Once we determine the value of the combined shapes, we then take the equation of centroid of x and y to plug in our value to received our final centroid value in x and y:

$$\bar{x} = \frac{420.35cm^2}{76.28cm} = 5.51cm$$

$$\bar{y} = \frac{373.48cm^2}{76.28cm} = 4.90cm$$

Interia (with respect to axes)

To find the inertial with respect to axes we took a shortcut that Dr. Zaurin.

We used the following formulas to find the inertia for the different shapes we have in the image.

$$I_{xcircle} = \frac{\pi r^4}{16}$$

$$I_{ycircle} = \frac{\pi r^4}{16}$$

Usually the equation above is just over 4, but since we have a quarter of a full circle we had to make it over 16.

$$I_{xrectangle} = \frac{bh^3}{3}$$

$$I_{yrectangle} = \frac{hb^3}{3}$$

These are your typical equations for centroid with rectangles, nothing changed

here.

Solving all these gets you the following values:

$$I_x \text{ circle} = 2874.75 \text{ cm}^4$$

$$I_y \text{ circle} = 2874.75 \text{ cm}^4$$

$$I_x \text{ rectangle} = 351.56 \text{ cm}^4$$

$$I_y \text{ rectangle} = 39.06 \text{ cm}^4$$

Now since the rectangle was cut out of the circle, we have to subtract the rectangle inertia from the circular one which gives us these final values for the inertia with respect to the axes:

$$I_x = 2523.19 \text{ cm}^4$$

$$I_y = 2835.67 \text{ cm}^4$$

Inertia (with respect to centroid)

We found the inertia with respect to the centroid in the same way that we did for the integral version above.

First recall that the centroid is located at $x=5.51\text{cm}$ and $y=4.90\text{cm}$

Also $I_x = 2523.19 \text{ cm}^4$ and $I_y = 2835.67 \text{ cm}^4$

We then found the area of the quarter circle and the rectangle, which we know what the formulas of those are.

We then subtract the area of the rectangle from that of the circle due to it being a cut out, and we got the following area value: 76.28 cm^2

Using these values, we can easily find the inertia from the centroid using the following formulas:

$$I_{ox} = I_x - A(\underline{y})^2$$

$$I_{oy} = I_y - A(\underline{x})^2$$

Then solving for those we get that:

$$I_{ox} = 691.71cm^4$$

$$I_{oy} = 519.80cm^4$$

Percent errors and potential error sources

For the percent errors of Centroids:

$$Percent\ Error = \frac{|measured - calculated|}{measured} * 100\%$$

Figure 1 (Integrals):

$$X = \frac{|0.4 - 0.375|}{0.4} * 100\% = 6.25\%$$

$$Y = \frac{|0.56 - 0.6|}{0.56} * 100\% = 7.14\%$$

Figure 2 (Composite Areas):

$$X = \frac{|5.5 - 5.51|}{5.5} * 100\% = 0.18\%$$

$$Y = \frac{|4.9 - 4.9|}{4.9} * 100\% = 0\%$$

While figure 2 had little to no error, figure 1 had some degree of error. This was most likely due to the fact that it was a parabolic shape; the shape of the curve was never constant, it was always changing. As a result, we struggled with trying to perfectly replicate figure 1 with $y = x^2$ as its shape.

Conclusion/discussion:

Vincent Tremblay (vi474918):

Overall I believe we did a great job. We were able to accurately compare the centroids from both the physical model and calculate it, which is always nice to do. It also balanced at the point we calculated which was nice to see.

With regards to the inertia, we were able to correctly find both for both models with respect to the axes and centroid.

Overall this was a great learning experience and we were able to successfully implement statics concepts.

Andrew Tran (an365172):

After completing this hands-on assignment, we became very familiar with the formulas used to find centroid and moment of inertia (respect to axes and centroid). The graphical and analytical portions for finding centroid was very successful in my opinion because it showed the accuracy of the formulas we

were using. The little percent error is evidence of how successfully concepts from statics can be easily applied to the real world.

Kevin Sanchez Ramos (ke987766):

In the end, after all our calculation, I felt like this was probably one of the unique hands on homework where we got a result that matched exactly what we physically measured. For instance, for the quarter of a circle we measured 5.5cm but in our hands on calculation we found that it was 5.51cm. This level of precision really shows how well we can calculate the centroid of shape despite part of the quarter of a circle being cut out and further shows our improvement in static work.